

## Multiple Choice

1. **Answer:** D

*Options:* A) BaH<sub>2</sub>; B) BaOH; C) Ba; D) BaO

*Note:* Correct option: D - BaO

2. **Answer:** D

*Options:* A) B; B) Al; C) In; D) Tl

*Note:* Correct option: D - Tl

3. **Answer:** D

*Options:* A) racemic mixture; B) single pure enantiomer; C) mixture of two diastereomers in equal amounts; D) meso compound

*Note:* Correct option: D - meso compound

4. **Answer:** B

*Options:* A) 0.0471 Hz; B) 21.2 Hz; C) 42.4 Hz; D) 66.7 Hz

*Note:* Correct option: B - 21.2 Hz

5. **Answer:** A

*Options:* A) hexane; B) 2-methylpentane; C) 3-methylpentane; D) 2,3-dimethylbutane

*Note:* Correct option: A - hexane

## True / False

6. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

7. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'Arsenic-doped silicon'.

9. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

## Long Form Response

11. **Answer:** detection of spin adducts requires lower power than direct detection of radical intermediates. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.  
*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:**  $r = k$ . Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.  
*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.